

Computer Science Department
University of Computer & Emerging Sciences (FAST-NUCES)

HOME WORK ASSIGNMENT COVER SHEET

COURSE TITLE	Software Design & Analysis	COURSE CODE	CS3004
INSTRUCTOR	DR. EIAZ AHMED	TYPE	☐ ☒ (Please tick)
		Indiv.	Group
ASSIGNMENT NO	HW-2 (Class)		
ASSIGNMENT	Use Case Diagram and Development of Template of POS		
HAND OUT DATE	25-SEP-2025	DUE DATE	05-OCT-2025
ESTIMATE TIME	8 Hours		

ASSESSMENT CRITERIA (or attached)	% Mark
Questions are attached (print copy) This HW includes the following tasks (Total 4) Requirement Analysis, Iterations, Use Case Diagram and Use Case Detailed Template Submission: Any mistake or missing material will lead to the reduction of 5% marks. Group of max 3 students is allowed, individual work is discouraged with 10% deduction	

TO BE COMPLETED BY STUDENT (TEAM LEAD)	GROUP MEMBERS ID						
NAME <u>Muhammad Moiz Ansari</u>	<table><tr><td>ID 23i-0523</td><td>Sec# F</td></tr><tr><td>ID 23i-0730</td><td>Sec# F</td></tr><tr><td>ID 23i-0528</td><td>Sec# F</td></tr></table>	ID 23i-0523	Sec# F	ID 23i-0730	Sec# F	ID 23i-0528	Sec# F
ID 23i-0523		Sec# F					
ID 23i-0730		Sec# F					
ID 23i-0528		Sec# F					
ID NO <u>23i-0523</u> Section# <u>F</u>							
Time Taken ~ 8 Hrs.							
DECLARATION: I/We declare that this Coursework is my/our group's own work							
SIGNATURES (All members)							

GRADE/ MARK AWARDED	☐	COMMENTS	
INSTRUCTOR'S SIGNATURE		DATE	

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REQUIREMENTS ANALYSIS

Functional Requirements

1. Manage Bills and Customers

- Generate bills for customers.
- Assign bill numbers automatically
- Capture necessary customer information (name, contact, etc.)

2. Item Search

- Search medicines by brand name or barcode number.
- Display medicine details.

3. Sales

- Add multiple items to the bill.
- Update item quantities.
- Apply discounts(if applicable).
- Handle different payment methods.
- Automatically calculate subtotal, discount, net total, and taxes.
- Print the customer invoice with the bill number, date/time, item details, customer details, total amount, discount, and balance.

4. Stock Handling

- Display current stock.
- Alert the cashier if the stock is insufficient.
- Prevent the sale of expired medicine.

Non-Functional Requirements

1. **USABILITY**: Simple UI for easy use and quick transactions.
2. **PERFORMANCE**: The System must generate invoices quickly.
3. **SCALABILITY**: Should handle multiple users without affecting performance.
4. **RELIABILITY**: The System should not lose data in case of a power failure.
5. **SECURITY**: Customer data and billing records must be secure.

ITERATIONS

1. Iteration 1: Sales Process

- Implement basic bill creation.
- Search/Select items.
- Add items to the cart.
- Calculate totals/discounts/taxes.
- Generate/Print invoices.
- Handle multiple payment methods.

2. Iteration 2: Customer Management

- Register customers by capturing their details.
- Store customers' past bills.
- Display customers' purchase history.

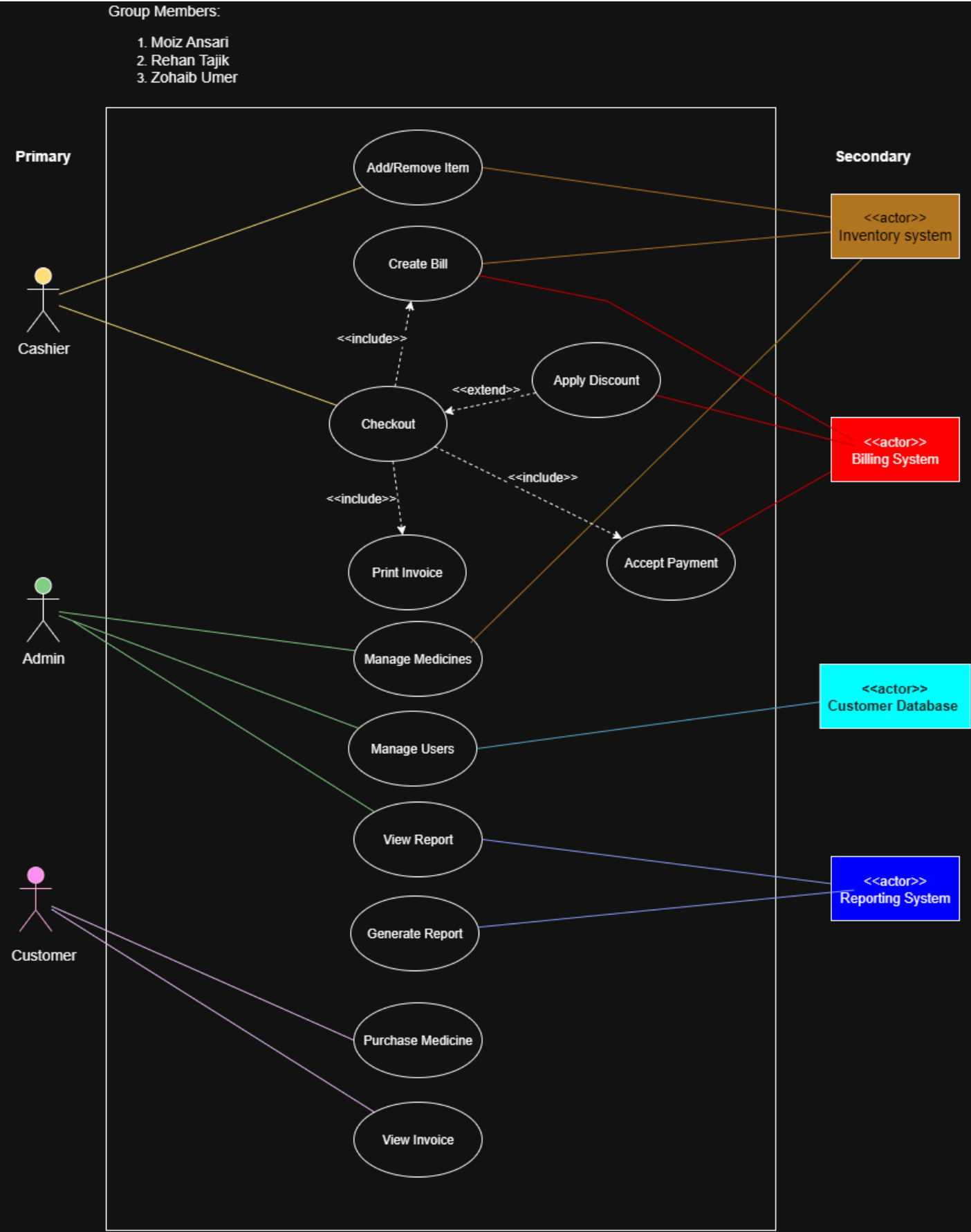
3. Iteration 3: Stock Handling

- Implement checks for stock availability.
- Update stock after each sale.
- Track expiry dates.

4. Iteration 4: Security

- Implement a login system for the cashier.
- Track cashier ID for each transaction.

Use Case Diagram



Welcome to MedPharma!

Bill#

Bill#

Customer Name

Enter Name...

Date

PriceTab

New

Medicine Name (Brand)

Medicine Id

Medicine Type

Quantity

T/P

Price

Dosage Description

MG ML

Qty/Item

Save

StockNo	Brand_name	Medicine_type	Quantity	Price_PerItem	Amount
3902850	CEDROX 1G 10S	Tablet	4	28.34	113.36
2309823	PANADOL 10G 20S	Tablet	3	100.0	300.0
4036742	ALLIUM CEPA 200	Drops	1	799.99	799.99

Stock#	Brand Name	Type
3902974	PANADOL SY...	Liquid
3902987	PANADOL DR...	Liquid
3902956	PANADOL TA...	Tablet
3902874	PANADOL EX...	Tablet

3

List

BarCode

Total

UserId/Login

Moiz Ansari

Max Disc. 15%

Status

Discount %age

Net Total

InvNo

Sale Mode

Discount AMT

Cash

StockQTY: 11752 ExpDate: 3/3/2018

Empty Dates

Calculate

Remove Med

Cancel

Balance

Print

Remove

Expiry D...

Stock#

Qty

Stock#

3/3/2018

39034...

117...

16

Print Receipt

MedPharma

123 Healthcare Ave, Wellness City

Tel: 555-MEDS (6357)

Date: 2024-05-20 Transaction # R123456789

ITEM	DESCRIPTION	QTY	QUANTITY	PRICE	TOTAL
1.	CEDROX 1G 10S	Tablet	4		113.36
2.	PANADOL 10G 20S	Tablet	3		100.00
3.	ALLIUM CEPA 200	Drops	1		799.99
Subtotal:					1213.35
Tax (8%):					97.07
Cash tendered: 13:40.00					Change: 39.58

Thank you for choosing MedPharma!